

Haesung Oh

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RESEARCH INTEREST

End-to-End Autonomous Driving

EDUCATION

Seoul National University, Interdisciplinary Program in AI, Master's Degree Feb 2024 - Current
Dynamic Robotic Systems Lab (DYROS Lab) Advisor: *Jaeheung Park*

POSTECH, Convergence IT Engineering, Bachelor's Degree Feb 2016 - Feb 2023
UC Berkeley, Exchange Student Jun 2017 - Dec 2017

Seoul Science Highschool Feb 2013 - Feb 2016

RESEARCH EXPERIENCE

Multi-Vehicle Zero-Shot Adaptation for E2E Autonomous Driving Advisor: *Jaeheung Park* Aug 2024 - Dec 2025

- Robust vehicle adaptation of an end-to-end autonomous driving AI model
- Physical dimensions are parameterized to predict corresponding waypoints for various vehicle models
- Achieved zero-shot adaptation and few-shot fine-tuning adaptation to unseen vehicle models
- *Keywords: End-to-end Autonomous Driving, Domain Adaptation, Vehicle Adaptation, CARLA*

Comfort Optimization for E2E Autonomous Driving using RL adapter Advisor: *Kichun Jo* Mar 2025 - Oct 2025

- Developed a PPO-based RL adapter to optimize the driving comfort of a baseline E2E planner within nuPlan simulation
- *Keywords: Reinforcement Learning, nuPlan, PPO, Driving Comfort*

MOTIE of RoK Gov - Alchemist Project: Next Generation Humanoid Advisor: *Jaeheung Park* Jul 2024 - Dec 2024

- Contribution to LLM-Humanoid connection for human-robot interaction
- *Keywords: LLM, Humanoid, Robotics, Control*

Metal-Organic Frameworks Property Regression (POSTECH iAI) Advisor: *Seungchul Lee* Sep 2022 - Dec 2022

- Development of a regression model for inverse design prediction of MOF properties
- Automatic hyperparameter tuning for optimized regression model performance
- *Keywords: Hyperparameter Tuning, Lazy Regression, Extra-tree Regression*

Autonomous Delivery Mobile Robot (SNU ARIL) Advisor: *Sungwoo Kim* Dec 2021 - Feb 2022

- Contribution to the autonomy simulation and optimization of a 6-wheel, skid steering autonomous delivery robot
- *Keywords: ROS, Unity simulation, Embedded system, 3D CAD, Hardware assembly*

Autonomous Parking Simulation Advisor: *Suhee Han* Sep 2021 - Dec 2021

- Development of an autonomous parking simulator using Unity ML-Agent and RL techniques
- Trained and evaluated the model with 8 million episodes for front, parallel, and diagonal parking scenarios, respectively
- *Keywords: Reinforcement learning, AI-based autonomy, Autonomy simulation, Unity ML-Agent*

Autonomous Drone Advisor: *Suhee Han* Feb 2021 - Jun 2021

- Implementation of an autonomous drone system using ultrasonic sensors to avoid obstacles and navigate terrain safely
- Rule-based approach utilized for extreme environment exploration based on obstacle conditions
- *Keywords: Rule-based autonomy, Arduino, Obstacle detection, PID control, Drone system dynamics*

Auto Training Routine Recording System Advisor: *Chulhong Kim* Sep 2020 - Dec 2020

- Development of a vision-based system to detect human pose and record weight training routines
- *Keywords: OpenPose, Pose estimation, Vector similarity*

Auto Strike Zone Detection Advisor: Sungjun Jung

Feb 2018 - Jun 2018

- Development of an automated system for real-time detection of strike zones in baseball
- *Keywords: Computer vision, Real-time, Line detection, Object detection*

WORKING EXPERIENCE

Alley 3D reconstruction AI model for fire trucks (Dudaji Start-up Company)

Apr 2023 - Aug 2023

- Development of an AI model that detects and reconstructs 3D objects in alleys
- The AI model determines whether fire trucks can safely navigate through the alleyways or not
- Integrated the model with a mapping server, allowing it to create blocker events and share them in real-time
- *Keywords: Deep neural network, Computer vision, 3D object detection, Monocular camera*

Autonomous Recycle Robot (Dudaji Start-up Company Internship)

Jul 2022 - Aug 2022

- Development of a 6-joint robot that classifies and perform pick-and-place tasks for recyclable items
- Contribution to item detection via camera, optimization of robot movement, and implement of pick-and-place operations
- *Keywords: Robotics, ROS, Computer vision, Yolo v5, Deep learning, Pick-and-place*

TEACHING EXPERIENCES

AI Instructor

Sep 2025 - Oct 2025

GenAI lecturer for Samsung Electronics DS Division

AI Instructor

Jul 2025 - Aug 2025

GenAI lecturer for Hansae Co., Ltd. Leadership Program in DX

AI Instructor

Jun 2025 - Jul 2025

GenAI lecturer for KakaoPay AI Capability-Building Workshop

AI Head Technical Coach

Jun 2025

2025 POSCO Group WX ZeroTon — led technical coaching for LLM-based projects

HONORS AND AWARDS

HOPE Scholarship

Dec 2024 - Dec 2027

Ilun Science & Technology Foundation

AI Fellowship Scholarship

Feb 2024 - Feb 2026

Seoul National University scholarship with Artificial Intelligence Graduate School Council

Creative ICT Scholarship

Feb 2016 - Dec 2021

MSIT (Ministry of Science and ICT), Korea, under the ICT Creative Consilience program

Korea Art Management Service: ART Hack-A-Thon

May 2018

Excellence Prize: *Art and coding education using drones for the next generation*

OTHER EXPERIENCE

Hyundai H-Mobility Class: Autonomous Car Class

Jul 2022 - Jan 2023

Keywords: Recognition, Decision making, Machine learning, Vehicle system control, Vehicle network

SKILLS

Programming Language Python, ROS, C/C++, CARLA, nuPlan, Git

2D/3D Graphic Unreal Engine, Unity, Fusion 360, AutoCAD